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Invention Conception Record

AgentForge: A Tiered AI Compute Token System for Query Complexity Control, Metered Consumption, and Environmentally Efficient AI Delivery

Inventor	John Kirshy
Employer	IBM
Conception Date	July 15, 2026
Record Created	July 15, 2026
Tools Used	IBM Bob (AI assistant) — IBM managed environment
Status	Pre-filing conception record — Provisional Patent Application pending
IBM IDF Filed	[ ] Pending submission to IBM IP portal

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1. INVENTION TITLE AND SUMMARY

A system and method for a **Tiered AI Compute Token Framework** that controls, prices, meters, and routes artificial intelligence query workloads across three defined complexity tiers. The system introduces a token-based economic model applicable to any large language model (LLM) or AI platform, enabling differentiated pricing, role-based access, consumption metering, enterprise licensing, and a structurally enforced reduction in unnecessary AI compute consumption.

2. PROBLEM BEING SOLVED

The inventor identified the following gaps in the current AI market:

- **Undifferentiated compute billing:** All AI queries — from "What is 2+2?" to "Build me a cloud infrastructure" — are treated and billed similarly, wasting energy and compute on simple tasks.
- **No structured complexity routing:** LLMs are invoked at full capacity regardless of query complexity, resulting in unnecessary data center load, power consumption, and cost.
- **Limited monetization structure:** AI application developers and LLM providers lack a standardized tiered pricing framework to monetize query complexity and user roles.

- **Environmental backlash:** The AI industry faces growing public and regulatory pressure over data center power consumption and carbon footprint, with no structural solution offered.
- **No metering layer:** No widely adopted system exists to measure, classify, and gate AI queries by complexity tier before execution, leaving consumption uncontrolled.

3. THE THREE TOKEN TIERS — CORE OF THE INVENTION

The inventor conceived a three-tier token classification system structured as follows:

Tier	Token Name	Use Case	Compute Level	Target User
1	Basic Token	Simple single-turn Q&A;, lookups	Low — lightweight model path	General / consumer users
2	Mid Token	Multi-turn sessions, explanations, analysis	Medium — standard LLM	Power users, SMB, analysts
3	Advanced Token	Complex research, infra generation (Azure/AWS), agentic pipelines	High — full agentic / large model	Developers, enterprise, architects

Each tier carries a distinct price point. Lower tiers are cheaper but consume faster when used for tasks beyond their design scope — creating a natural economic incentive for users to self-select the appropriate tier. Developers and power users purchase Tier 3 tokens; general consumers purchase Tier 1 or Tier 2 tokens.

4. THE TOKEN METERING LAYER

The inventor conceived a **Token Metering Engine** — a measurement and enforcement layer positioned between the user request and the LLM execution environment. This layer:

- **Classifies** incoming queries against the purchased token tier.
- **Gates or allows** execution based on whether the user's token tier covers the query complexity.
- **Logs and ledgers** consumption against the user or enterprise token balance in real time.
- **Routes** queries to the appropriate model path (lightweight vs. full LLM vs. agentic pipeline) based on tier.
- **Produces** an auditable consumption record — useful for enterprise compliance, billing, and environmental reporting.

> **Note:** The specific technical implementation of the metering layer is intentionally not detailed here in order to preserve patentability. The concept of the metering layer as a system component is the inventor's original idea, conceived independently on July 15, 2026.

5. LICENSING AND REVENUE MODEL

The inventor identified three distinct markets for the tiered token system:

- **Consumer app market:** AI-powered applications (video generators, voiceover tools, coding assistants, creative tools) sell tiered token credits directly to end users. Tier 1 is the entry product; Tier 3 is premium. Users buy more tokens as they need more capability.
- **Enterprise licensing:** Large organizations license the tier framework on a per-seat or departmental basis. Developers receive Tier 3 access; general staff receive Tier 1 or 2. Pricing scales by tier and volume.
- **LLM platform licensing:** The tier system is offered to AI companies (any LLM provider) as a standardized monetization and consumption-control framework — enabling them to offer differentiated products, attract new customer segments, and credibly market a green AI positioning backed by structural enforcement.

6. ENVIRONMENTAL CLAIM

By routing simple queries to Tier 1 — which never invokes full LLM or agentic infrastructure — the system structurally enforces compute efficiency. The metering layer prevents unnecessary high-compute invocations. The result is a measurable, auditable reduction in data center load, power consumption, and cooling demand compared to undifferentiated AI query systems.

This gives AI companies and enterprises a credible, architecturally-backed environmental claim — not greenwashing, but a structural property of the system design.

7. AGENTIC AI CONNECTION

The inventor noted that Agentic AI — where queries are structured around defined tasks and patterns rather than open-ended prompts — naturally aligns with the Tier 3 design. Agentic structure constrains what the LLM must compute, causing the model to work less hard than it would for an unbounded generative prompt of equivalent apparent complexity. This is an additional environmental efficiency benefit of the tiered agentic approach.

8. CONCEPTION RECORD — EVIDENCE STATEMENT

> The inventor, John Kirshy, conceived all ideas described in this document independently during a conversation conducted on **July 15, 2026** using IBM Bob, an IBM-managed AI assistant. The conversation record is preserved within IBM's platform and constitutes timestamped evidence of the conception date. The inventor articulated every concept; IBM Bob served only as a clarifying and recording tool. No prior art was presented to the inventor during conception.

Inventor Signature	John Kirshy ______________________________

Date	July 15, 2026
Witness / Notary	________________________________
IBM IP Coordinator	________________________________
IDF Reference #	[To be assigned upon IBM IDF submission]